

Web Development Crash Course

**React • JavaScript • MySQL
CSS • Tailwind CSS • Bootstrap**

21 September - 31 December 2026

A practical training plan for a junior IT employee, supervised by the Head of IT. React is a required outcome of this programme.

Duration	Weekly commitment	Final outcome
14 teaching weeks + final review	15 hours / 3 hours per working day	A React application connected to MySQL through an API

Training approach

Build one increasingly capable IT Asset Tracker throughout the course. Learn HTML and CSS first, establish JavaScript foundations, then develop the React interface, database and backend connection.

Complete a separate Bootstrap dashboard in Week 2. Use Tailwind for the React project from Week 8 onward to keep its styling consistent.

Node.js and Express provide the small backend needed to connect React to MySQL. Database credentials stay on the server; the browser communicates with the API.

Scope and expectations

Assumes beginner-level development experience and basic laptop skills. The target is supervised contribution to web projects. Authentication and production deployment are outside the core scope. Use fictional employees and equipment.

Final review: 28-31 December

Correct outstanding issues, implement a small feature and diagnose one bug independently. Assess understanding as well as the finished application.

Laptop tools and weekly routine

Tool / download link	Purpose
VS Code	Code editor
Node.js LTS + npm	JavaScript tools and backend runtime
Git	Track code changes
GitHub	Assignments and code reviews
Chrome DevTools	Inspect styles, debug code and requests
MySQL Community Server	Local database
MySQL Workbench	Tables, SQL queries and data inspection
Postman	Test API requests
Vite	Create and run the React project

Install Bootstrap, Tailwind and Express within the relevant projects using their official setup guides. Select installers compatible with the laptop operating system. The Head of IT should assist with initial configuration.

Weekly time allocation

Activity	Hours
Tutorials and guided exercises	3
Build the assignment	9
Test, debug and document	2
Demonstration and review with Head of IT	1

Every week: submit code to GitHub, demonstrate the assignment, explain a function or component, describe one diagnosed bug and implement a small change live. Record unfinished work as GitHub Issues.

AI may provide explanations, hints and feedback. The employee must understand every submitted line; the live assessment should be completed without AI assistance.

Foundations and responsive design

WEEK 1 | 21-27 Sep 2026

HTML, CSS and setup

Study: Page structure, labelled forms, selectors, box model, spacing and typography. Set up the tools and practise a Git commit.

Submit: Build a two-page IT services website with a support-request form. Push the source to GitHub.

Learn: [HTML](#) / [CSS](#) / [Git](#)

WEEK 2 | 28 Sep-4 Oct 2026

Responsive CSS and Bootstrap

Study: Flexbox, Grid, media queries, Bootstrap layout, utilities, navigation and forms.

Submit: Make the website responsive. Build a separate Bootstrap dashboard with cards, a table and a form.

Learn: [CSS](#) / [Bootstrap](#)

WEEK 3 | 5-11 Oct 2026

JavaScript fundamentals

Study: Variables, data types, operators, conditions, loops, functions and scope.

Submit: Complete 15 small exercises and build an equipment-cost calculator.

Learn: [JavaScript](#)

JavaScript to React

WEEK 4 | 12-18 Oct 2026

JavaScript data and interaction

Study: Arrays, objects, DOM selection, events, validation, map and filter.

Submit: Build an equipment list with add, delete, search and filter functions.

Learn: [JavaScript](#)

WEEK 5 | 19-25 Oct 2026

JavaScript for React

Study: Modules, destructuring, spread syntax, arrow functions, promises, async/await, Fetch and JSON.

Submit: Load sample JSON through the local server. Display loading and error states and explain the code during review.

Learn: [JavaScript](#) / [Fetch](#)

WEEK 6 | 26 Oct-1 Nov 2026

React fundamentals

Study: Vite setup, JSX, components, props, lists, keys and event handlers.

Submit: Rebuild the equipment list with reusable React components.

Learn: [React](#) / [W3Schools React](#)

React and Tailwind

WEEK 7 | 2-8 Nov 2026

React state and forms

Study: useState, controlled inputs, validation, immutable updates and lifting state.

Submit: Implement create, edit, delete, search and filter features using React state.

Learn: [React state](#)

WEEK 8 | 9-15 Nov 2026

Tailwind CSS

Study: Vite integration, utility classes, responsive layout, spacing, typography, hover and focus states.

Submit: Style the React application with Tailwind. Build reusable buttons, inputs and cards for mobile and desktop.

Learn: [Tailwind setup](#) / [Tailwind utilities](#)

WEEK 9 | 16-22 Nov 2026

React integration

Study: Effects, fetching data, loading/error/empty states, effect cleanup and basic routing.

Submit: Create Dashboard, Assets and Asset Details pages. Fetch sample data and handle failed requests.

Learn: [React effects](#) / [React Router](#)

MySQL and the backend

WEEK 10 | 23-29 Nov 2026

MySQL fundamentals

Study: Databases, tables, data types, primary keys, SELECT, INSERT, UPDATE, DELETE, filtering and sorting.

Submit: Create an asset database with at least 20 fictional records. Submit 15 SQL queries.

Learn: [MySQL](#)

WEEK 11 | 30 Nov-6 Dec 2026

Relational database design

Study: Foreign keys, relationships, joins, constraints, grouping and basic indexes.

Submit: Create related assets, categories and employees tables. Query assignments and counts by category.

Learn: [MySQL](#) / [MySQL docs](#)

WEEK 12 | 7-13 Dec 2026

Node.js and Express API

Study: HTTP methods, routes, JSON, status codes, environment variables, server validation and parameterised SQL.

Submit: Build API endpoints to list, create, update and delete assets in MySQL. Test valid and invalid requests.

Learn: [Node.js](#) / [Express](#) / [MySQL2](#)

Integration and final delivery

WEEK 13 | 14-20 Dec 2026

Full application integration

Study: Connect React screens and forms to the Express API and MySQL. Manage success and failure feedback.

Submit: Finish the IT Asset Tracker. Changes must persist in MySQL and display correctly after browser refresh.

Learn: [React](#) / [Fetch](#) / [Express](#) / [MySQL](#)

WEEK 14 | 21-27 Dec 2026

Testing and handover

Study: Responsive checks, keyboard use, API failures, database constraints, code cleanup and documentation.

Submit: Submit the application, database setup script, README and test checklist. Give a 15-minute demonstration.

Learn: [DevTools](#) / [GitHub](#)

FINAL REVIEW | 28-31 Dec 2026

Independent assessment

Resolve review issues, implement one small requested feature and diagnose one bug without AI assistance. Explain the flow from React to the API to MySQL and back.

Required handover

Source repository; database schema and fictional seed data; setup instructions; environment-variable example without secrets; test checklist; known limitations; and a 15-minute demonstration.

Final project and assessment

IT Asset Tracker

Build a React and Tailwind interface with Dashboard, Assets and Asset Details pages, backed by Node.js/Express and MySQL.

- Create, view, edit and delete equipment records.
- Store unique asset ID, name, category, assigned employee and status.
- Search assets and filter by category and status.
- Show total, assigned and available asset counts.
- Populate employee and category choices from MySQL records.
- Validate required fields in the interface and on the server; enforce unique asset IDs in the database.
- Use parameterised SQL and keep database credentials on the server.
- Show loading, empty, success and error states; retain changes after refresh.
- Support mobile and desktop layouts, keyboard use and labelled forms.

Assessment rubric

Area	Weight
Working required features and MySQL persistence	35%
Explains and modifies own React / JavaScript code	25%
Responsive layout and accessibility	15%
Debugging, API checks and testing	15%
Git usage and documentation	10%

Pass mark: 75%. Working core features and the ability to explain the code are mandatory. If progress slips, reduce optional polish and use the final review window for corrections; retain React and MySQL as required outcomes.

Supervisor checkpoints: Week 5 - JavaScript readiness; Week 9 - React and Tailwind demonstration; Week 12 - working API and SQL; Week 14 - complete application.